University of Burgundy

Polytech Dijon

Path Planning with Kinematic Constraints

Bicycle Model and RRT Extensions

Project Repository:
`github.com/AdhamElkomi/PRM_RRT_Project`

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Introduction

In this part of the project, we focused on path planning under kinematic constraints. Unlike the previous part where the robot could move freely in straight lines, here we introduced a bicycle model to simulate a more realistic motion.

The main idea is that a real vehicle cannot move in any direction instantly. Its motion depends on its orientation and steering angle, which makes the problem more constrained.

The goal of this work was therefore to understand how these constraints affect path planning and how classical algorithms like RRT can be adapted to handle them.

Exercise 1: Bicycle Model

We implemented a bicycle model using Euler integration. The state of the system is defined by position, orientation and steering angle.

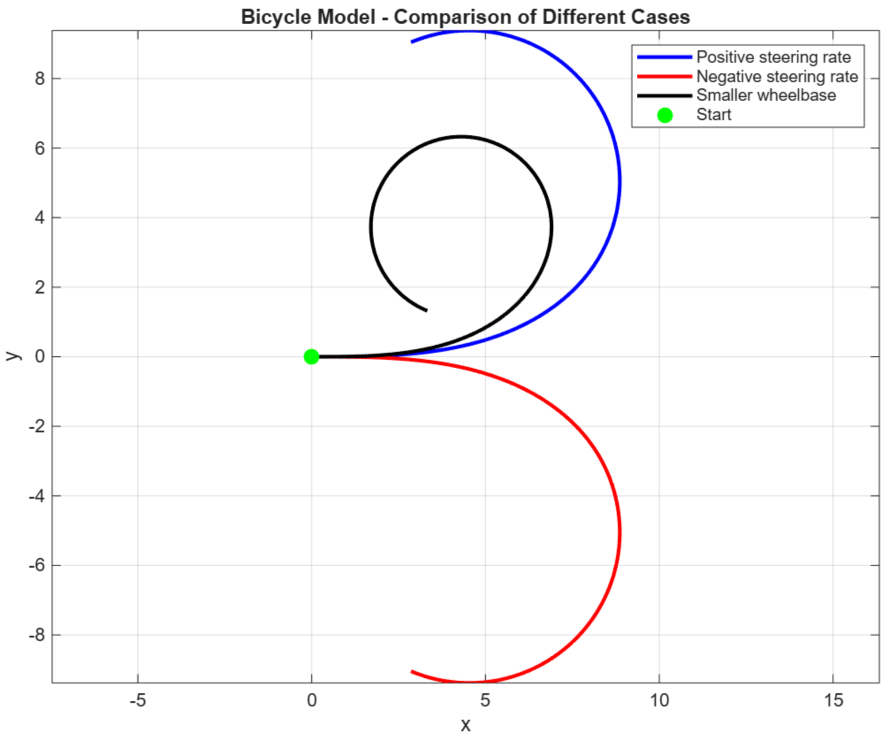

Figure 1: Bicycle model behavior under different steering conditions

By testing different steering inputs, we can clearly observe the effect of the model.

When the steering rate is positive, the vehicle turns in one direction, while a negative steering rate produces a turn in the opposite direction. We also tested the influence of the wheelbase: a smaller wheelbase leads to tighter curves, which makes the vehicle more reactive.

These results confirm that the model behaves as expected and is suitable to simulate realistic motion.

Exercise 2: Kinematic RRT

In this part, we modified the classical RRT algorithm to take into account the bicycle model.

Instead of connecting nodes with straight lines, we now generate small feasible trajectories using the model. Each expansion of the tree corresponds to a motion that could actually be executed by the vehicle.

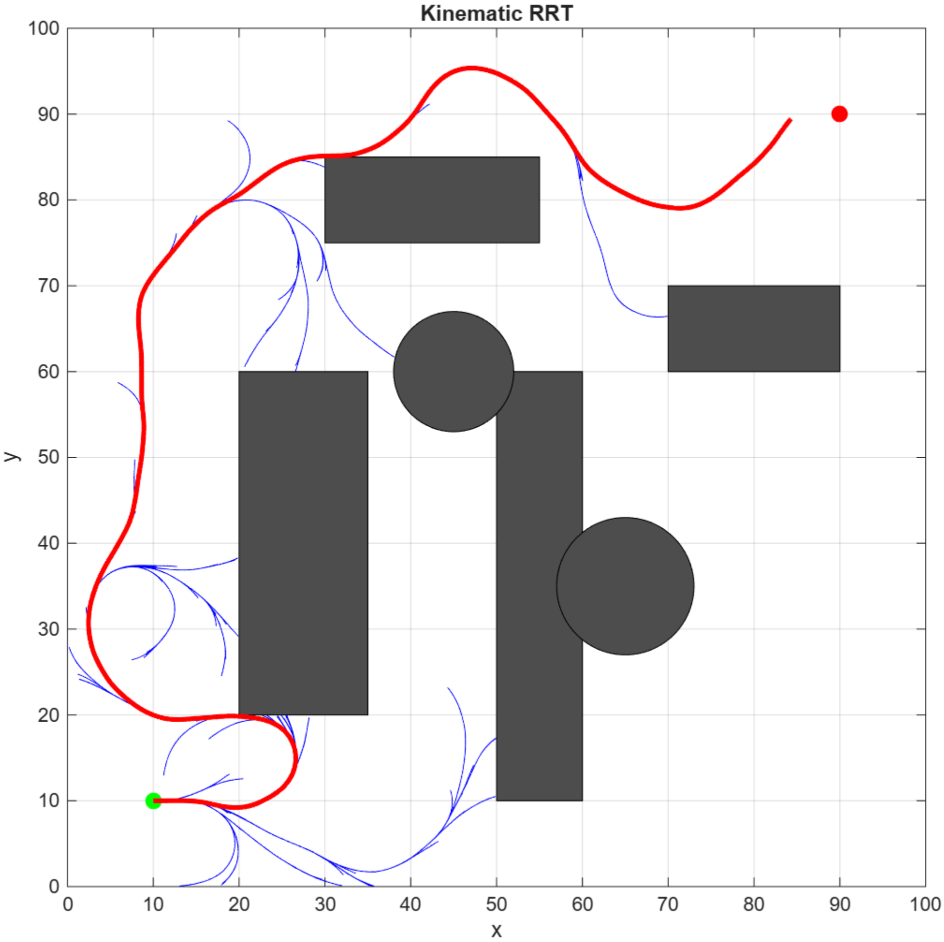

Figure 2: Kinematic RRT

The main difference is clearly visible: the tree is no longer composed of straight segments, but of smooth curves. This makes the solution more realistic, even if the algorithm becomes slightly slower and more complex.

The resulting path follows natural vehicle motion, which is the main objective of this modification.

Exercise 3: Feasible Kinematic RRT

In this exercise, we improved the previous approach by modifying how the nearest node is selected.

Instead of simply choosing the closest node in terms of Euclidean distance, we consider which node can actually reach the sampled point more effectively using feasible trajectories.

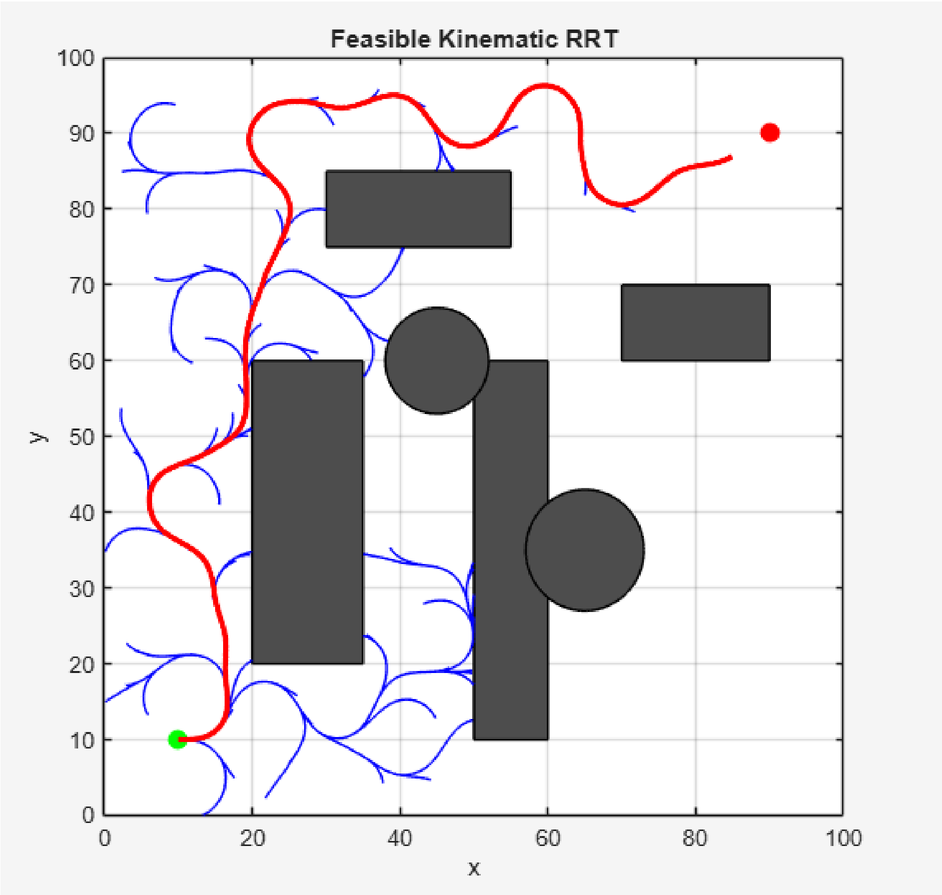

Figure 3: Feasible Kinematic RRT

This change improves the exploration of the space. The tree becomes more structured and avoids useless expansions.

As a result, the algorithm tends to find better paths with fewer iterations. This shows that taking into account the reachable set is more relevant than using simple geometric distance.

Exercise 4: Path Improvement

Finally, we applied a simple post-processing step to improve the path obtained from RRT.

The idea is to remove unnecessary intermediate points by checking if two distant points can be connected directly without collision.

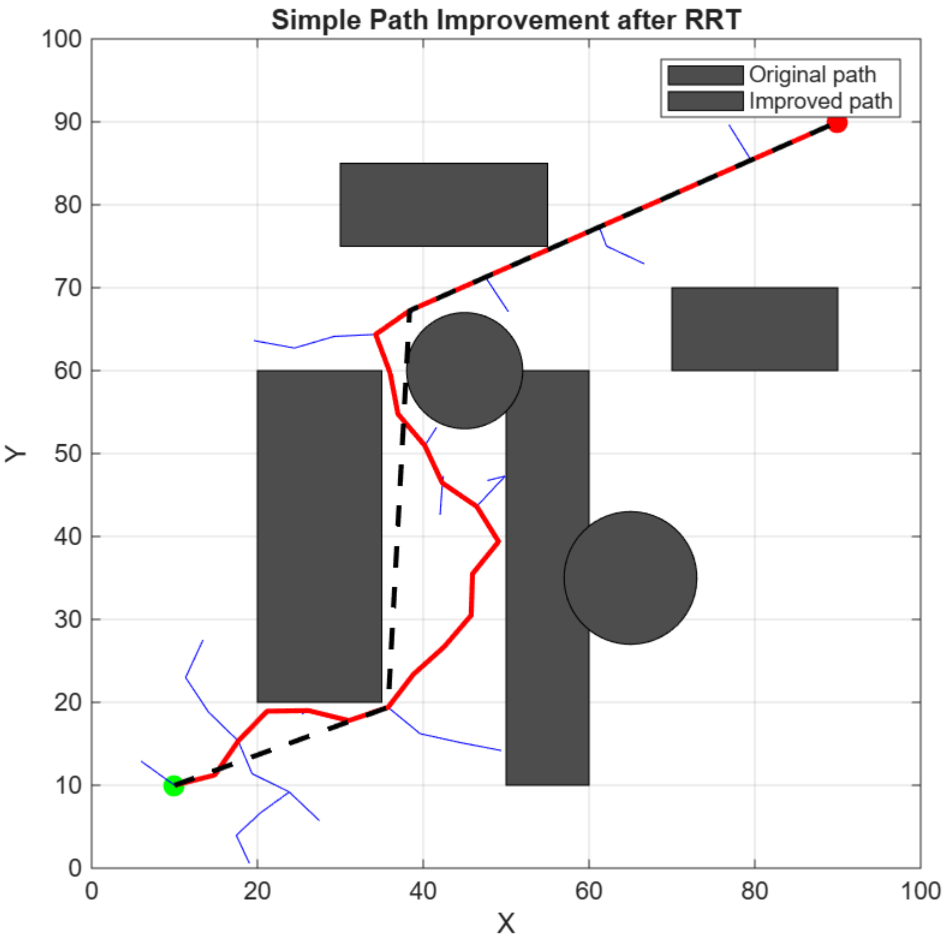

Figure 4: Path improvement after RRT

We can clearly see that the improved path is shorter and more direct. Some detours created by the random exploration are removed.

This step is simple but very effective. It allows us to improve the quality of the solution without modifying the core algorithm.

Conclusion

This work shows the importance of considering kinematic constraints in path planning.

The bicycle model allows more realistic motion, but increases complexity. The improved RRT strategies help to maintain efficiency while producing better trajectories.

Overall, combining kinematic modeling with smarter exploration leads to more practical and realistic path planning.